Samrath Bawa

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Skills

Programming: C++, Java, Python, R, JavaScript, HTML, CSS, x86 Assembly, Git, JUnit, REST APIs, pandas
Embedded and Electrical: Teensy, ESP32, CAN bus, PlatformIO, Altium Designer, KiCad, Soldering, Lighting
CAD and Analysis: SolidWorks, Fusion 360, Ansys Mechanical (FEA)
Fabrication: 3D printing (machine upkeep, slicing), CO2 laser cutting and engraving, woodworking
Game Engines: Godot, Defold, Panda3D

Experience

AI Training Fellow July 2026 – Present
Handshake AI *Remote*

- Writing hard questions about charts, tables, and images built to find where AI models break, tagged by the kind of reasoning they test
- Writing money and tax questions that can only be answered by reading through five or more documents
- Building grading rubrics that score AI answers point by point and writing up what each answer got wrong, which becomes training data
- Rating and captioning AI generated video, images, and audio against a detailed style guide

Software Captain 2026 – Present
Bronco Electric Motorsports (Formula SAE Electric) *Pomona, CA*

- Lead the software side of the electric race car and wrote the team's code, Git, and docs standards
- Responsible for the car's CAN bus network, covering the dashboard, wheel display, and sensor board
- Managed other members' code contributions and reviewed their pull requests to ensure they met the team's standards
- Refactored the full firmware codebase of last year's car into modules to build the new car on

Vice President August 2023 – June 2025
Woodbridge High School Web Development Club

- Taught members HTML, CSS, and JavaScript and built websites for clients such as other clubs and teachers

Jr. Auxiliary Volunteer May 2024 – May 2025
Hoag Hospital

- Supported patients, visitors, and staff with wayfinding and wheelchair services

Projects

Race Car Dashboard and Steering Wheel Display | *Teensy, CAN bus, Altium*

- Designed a sunlight readable CAN wheel display and a Teensy 4.1 dashboard schematic for the car
- Moved the team's ESP32 sensor board from KiCad to Altium, verifying all 110 parts and 75 nets

Accelerator Pedal Stress Analysis | *SolidWorks, Ansys*

- Ran FEA on the pedal shock mounts, found the stock part too close to failing, and designed an aluminum pedal stop (safety factor 2.4)
- Built a posable 95th percentile driver model in SolidWorks to check the car against the FSAE driver fit rules

Game Telemetry to Hardware Bridge | *Windows, Device Drivers*

- Turned live car data from a racing game into steering forces on a real steering wheel, using virtual device drivers

Education

California State Polytechnic University, Pomona Pomona, CA
Bachelor of Science in Computer Science, GPA 3.61, Dean's List *2025 – 2029*

- Coursework: Data Structures, Computer Organization and Assembly, Systems Programming, C++, Discrete Structures, UI Design, Probability and Statistics

Woodbridge High School
High School Diploma *2021 – 2025*